

Learning What Can Combine and What Can Substitute: Relational Inductive Biases for BabyLM Pretraining

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Abstract

Language can produce many expressions from a limited set of units because linguistic units participate in two basic relations. Syntagmatic relations determine which units can combine, while paradigmatic relations determine which units can substitute for one another in the same position. We study whether explicitly learning these relations helps language models generalize from limited data. We introduce two auxiliary objectives for masked language modeling. The Syntagmatic objective aligns a masked representation with relevant context selected by attention, while the Paradigmatic objective aligns it with the embedding neighbors of the target token. Both objectives are generated by the model itself and require no additional data, annotation, or trainable parameters. Experiments on BabyLM 2026 Strict-Small show that both single-relation systems improve mean Overall performance over Standard MLM, with Paradigmatic giving the most consistent gains across three seeds. Relay Joint combines the two objectives and shows a strong neighborhood-alignment signal at its submitted checkpoint, although its downstream gains vary across seeds. These results suggest that relational objectives can provide useful learning signals under limited data, while stronger internal alignment does not always lead to more stable downstream performance. Code is now publicly available on [GitHub](#).

1 Introduction

Language is an economical, efficient, and flexible hierarchical symbolic system. A finite inventory of linguistic units supports a virtually unbounded range of expressions because these units are not isolated: they participate together in syntagmatic relations of combination and paradigmatic relations of choice. Syntagmatic relations determine which units can combine to form larger structures (Sausure, 1916; Firth, 1957), whereas paradigmatic relations organize the alternatives that may occupy

the same structural position (Sausure, 1916; Halliday, 1961; Harris, 1954). Together, these axes support the hierarchical organization, productivity, and generalization capacity of language.

This organization suggests a hypothesis for learning from limited input. Learners cannot memorize every possible expression; they must abstract structures and categories that transfer beyond observed instances (Tomasello, 2003; Bybee, 2010). Combination relations provide regularities over how units form larger patterns, while choice relations organize categories of fillers for a shared position. If a language model is explicitly encouraged to capture both kinds of regularity, it may rely less on memorizing individual expressions and instead form reusable structures and categories. We refer to this proposal as the **Relational Efficiency Hypothesis**. It is a research hypothesis about fixed-exposure generalization, not a claim of computational speed.

Masked language modeling (MLM) (Devlin et al., 2019) already offers implicit opportunities to learn both relations: predicting a missing token requires sensitivity to context and lexical alternatives. However, token-level cross-entropy directly supervises only the identity of the ground truth token. It does not explicitly require the masked representation to retain which contextual units support combination at that position or which lexical units approximate a potential choice class. We do not assume that standard MLM lacks these properties; rather, we test whether explicit relational representation targets provide a useful inductive bias when input is limited (Warstadt et al., 2023; Hu et al., 2024).

We operationalize the two axes through detached, model-internal targets at MLM-selected positions. Syntagmatic uses last-layer attention to construct an attention-weighted target from eligible visible context, approximating contextual support. Paradigmatic uses the current input-embedding neighbors of the ground truth token to construct

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a neighborhood centroid, approximating substitutional choice. Neither objective adds corpus data, external annotation, or trainable parameters. We compare the two single-relation systems and Relay Joint across seeds 0, 1, and 42 under the same backbone, corpus, masking process, and exposure budget. Paradigmatic achieves the highest and most consistent mean Overall result. The submitted Relay checkpoint is the strongest individual checkpoint among the four primary systems, but its advantage does not persist across seeds. The results are therefore task- and seed-dependent and do not establish universal synergy.

Our contributions are:

1. **Theory.** The Relational Efficiency Hypothesis connects syntagmatic–paradigmatic organization to language generalization under limited exposure.
2. **Method.** Two self-generated, detached, and parameter-neutral relational targets approximate contextual combination and substitutional choice at MLM-selected positions, together with the staged Relay Joint schedule used by the submitted checkpoint.
3. **Evidence.** We provide a three-seed, seed-matched downstream comparison, together with matched checkpoint trajectories for seed 0 and a 2,000-item representation analysis over 31 target-token-ID clusters. The latter uses clustered-bootstrap uncertainty and should not be interpreted as across-seed variance.

2 Related Work

2.1 Syntagmatic and Paradigmatic Organization

Saussure (1916) distinguished syntagmatic relations among units combined in sequence from associative, later paradigmatic, relations among alternatives *in absentia*. Halliday (1961) recast this contrast as structure (chain) and system (choice), while Harris (1954) used distributional equivalence to characterize substitution classes. Sahlgren (2008) connected the distinction to distributional representation: direct co-occurrence tends to emphasize syntagmatic association, whereas similarity of contexts tends to emphasize paradigmatic association.

Usage-based and construction-based theories further explain how repeated instances can

support schemas with structurally constrained slots (Tomasello, 2003; Bybee, 2010; Goldberg, 2006). Word-association studies document related contrasts in human responses (Brown and Berko, 1960; Ervin, 1961; Entwisle et al., 1964). These traditions motivate the relational axes, but our continuous targets are model-internal approximations rather than symbolic linguistic structures. The unresolved question is whether both axes can be operationalized as useful inductive biases for pretraining from limited exposure.

2.2 Relational Signals in Language Models

Sun et al. (2015) modeled syntagmatic and paradigmatic relations in static embeddings, while Mickus (2024) interpreted MLM through the paradigmatic axis. Lexical-substitution work instead generates or ranks context-appropriate substitutes with contextual models (Melamud et al., 2016; Zhou et al., 2019) or human annotations (Lee et al., 2021). Our objective neither generates substitutes nor uses labeled substitute sets; it treats the current input-embedding neighborhood as a distributional approximation.

Attention analyses provide another source of relational evidence: individual heads can correlate with syntactic relations (Clark et al., 2019), although attention is not a guaranteed explanation or syntactic ground truth (Jain and Wallace, 2019). Similarly, distributional co-occurrence explains much but not all MLM generalization (Sinha et al., 2021; Chiang and Yogatama, 2023). Unlike post-hoc analyses, both objectives here are generated from internal model signals and train MLM-selected contextual hidden states. The gap we address is whether these approximate signals can serve as training biases rather than merely describe representations after training.

2.3 Data-Constrained Pretraining

The BabyLM challenge (Warstadt et al., 2023; Hu et al., 2024; Charpentier et al., 2025) tests language learning under tightly constrained input. Existing work has improved such training through architectural changes (Huebner et al., 2021), alternative or combined training objectives (Charpentier and Samuel, 2024), adaptive masking strategies (Edman and Fraser, 2025), and data organization (Haga et al., 2024). These approaches show several routes to stronger limited-data models, but do not isolate the effect of the two relational axes on the representation at a masked position. Our comparison there-

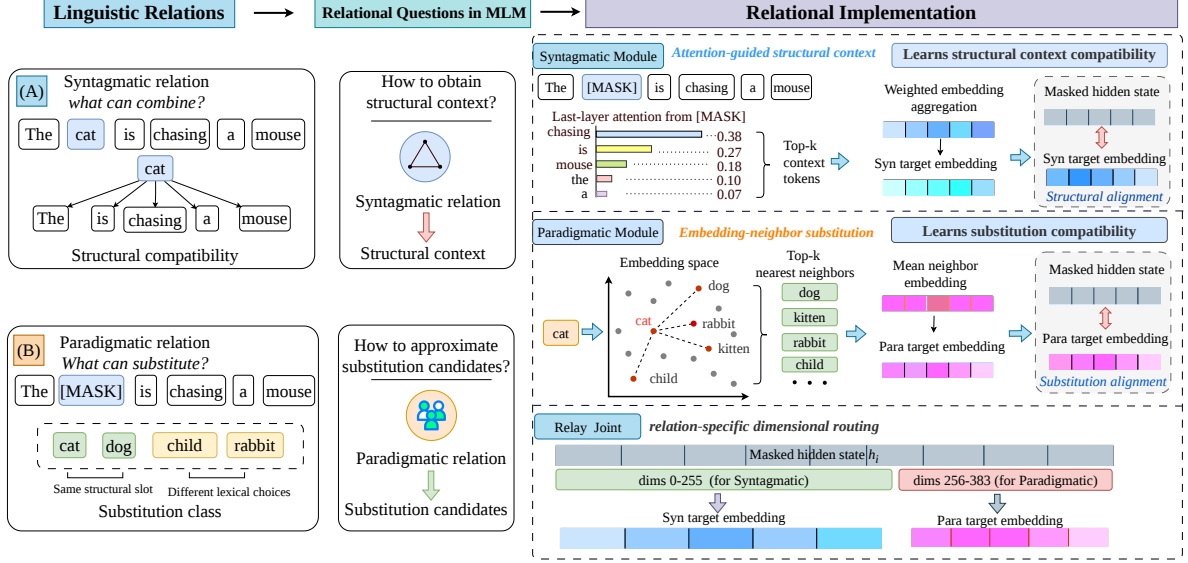


Figure 1: Overview of the relational pretraining framework. The four systems share the backbone, corpus, MLM corruption process, optimizer, and exposure budget, while differing in their relational objectives and how these objectives are integrated during training.

fore fixes the corpus, corruption process, backbone, and cumulative exposure, changing only the detached relational targets applied to MLM-selected hidden states.

3 Method

Let x_i be the ground truth token at an MLM-selected position i , and let $\mathbf{h}_i \in \mathbb{R}^{384}$ be its final-layer contextual hidden state before the MLM prediction head. Standard MLM minimizes token cross-entropy $\mathcal{L}_{\text{MLM}}$. Both relational objectives construct a target from signals already available inside the model and apply stop-gradient to that target. Consequently, an auxiliary target cannot move simply to reduce its own loss; the relational gradient updates the model through the contextual hidden-state branch. Neither objective requires new training data or introduces additional trainable parameters.

3.1 Syntagmatic Alignment

Syntagmatic relations concern how linguistic units combine within an expression. We approximate this relation by asking which visible context positions the model’s final attention layer treats as support for an MLM-selected position. We average the last-layer attention weights across all 12 heads. After excluding the target position, padding, $[\text{CLS}]$, $[\text{SEP}]$, $[\text{MASK}]$, and every other MLM-selected position, we select the top $K = 8$ remaining posi-

tions. Let $\mathcal{A}_i$ denote this set. We apply a softmax to the selected attention values to obtain normalized weights $\tilde{\alpha}_{ij}$ over $\mathcal{A}_i$. Using embeddings of the original, uncorrupted input tokens, the target and loss are

$$\mathbf{s}_i = \text{sg} \left(\sum_{j \in \mathcal{A}_i} \tilde{\alpha}_{ij} \mathbf{e}(x_j) \right), \quad (1)$$

$$\ell_{\text{syn},i} = 1 - \cos(\mathbf{h}_i, \mathbf{s}_i).$$

where sg denotes stop-gradient.

The loss acts on the complete 384-dimensional hidden state. It is weighted by $\lambda_{\text{syn}} = 5 \times 10^{-4}$ and becomes active after approximately 10% of training. MLM-selected entity tokens and tokens deemed invalid by the fixed vocabulary lookup do not receive this target. This target approximates prediction-relevant contextual support; it is not an explicit dependency representation.

3.2 Paradigmatic Alignment

Paradigmatic relations concern the alternatives that can occupy a shared structural position. Because the training data provide no explicit substitution sets, we approximate this relation using the ground truth token’s neighborhood in the model’s current input embedding space. For each eligible MLM-selected position, we select the $K = 8$ vocabulary items with the highest cosine similarity to the ground truth token in the current input-embedding

space, excluding the ground truth token itself. We denote this set by $\mathcal{N}_i$.

The target and loss are

$$\mathbf{p}_i = \text{sg} \left(\frac{1}{K} \sum_{v \in \mathcal{N}_i} \mathbf{e}(v) \right), \quad (2)$$

$$\ell_{\text{para},i} = 1 - \cos(\mathbf{h}_i, \mathbf{p}_i).$$

The neighbor centroid is aligned with the full 384-dimensional hidden state; no content projection is used. For syntax tokens, the loss weight is 5×10^{-5} and activation begins after approximately 55% of training. For content tokens, the weight is 10^{-6} and activation begins at 99%, so the loss is active only during the final 1%. Entity and reading-sensitive tokens do not receive a Paradigmatic loss. The fixed vocabulary categories and their lookup priority are detailed in Appendix A. Embedding neighborhoods are distributional approximations to potential alternatives, not guaranteed grammatical substitutes in a particular context.

3.3 Relay Joint Objective

The formal submission model combines the two targets with a staged, fixed-subspace schedule. The Syntagmatic loss is applied to dimensions 0–255 of the contextual state and target, while the Paradigmatic loss is applied to dimensions 256–383. These are fixed coordinate slices, not learned projections. Standard MLM continues to use the complete hidden state, and the auxiliary design adds no trainable parameters.

Syntagmatic supervision begins after approximately 10% of optimization with weight 5×10^{-4} . Syntax-token Paradigmatic supervision begins after approximately 20% with weight 5×10^{-5} . From that point onward, the Syntagmatic per-position weight on syntax tokens is retained at 10% of its earlier strength rather than removed. Content-token Paradigmatic supervision has weight 10^{-6} and begins at 99%, so it is active only in the final 1% of training. Entity and reading-sensitive tokens receive no Paradigmatic loss, and entity tokens receive no Syntagmatic loss.

Relay Joint also applies an entity representation anchor to eligible MLM-selected entity tokens from approximately 10% to 60% of training, using the same cosine-alignment form, $1 - \cos(\mathbf{h}_i, \text{sg}(\mathbf{e}(x_i)))$, between the contextual state and the detached ground truth token embedding.

Method	Syn Para		Rel. rep.	Ent. anchor
Standard MLM	No	No	–	No
Syntagmatic	Yes	No	Full 384	No
Paradigmatic	No	Yes	Full 384	No
Relay Joint	Yes	Yes	Syn 256 / Para 128	Yes

Table 1: The four primary systems. Relay Joint uses fixed slices 0–255 for Syntagmatic alignment and 256–383 for Paradigmatic alignment; MLM prediction always uses the full state.

The anchor uses the complete 384-dimensional hidden state and has weight 2.5×10^{-4} .

In Relay Joint, $\mathcal{L}_{\text{syn}}$ uses dimensions 0–255 and both Paradigmatic losses use dimensions 256–383. Each loss is averaged over eligible positions; $g(t)$ is its progress gate. The Relay Joint objective is

$$\begin{aligned} \mathcal{L}(t) = & \mathcal{L}_{\text{MLM}} + \lambda_{\text{syn}} g_{\text{syn}}(t) \mathcal{L}_{\text{syn}} \\ & + \lambda_{\text{para,syn}} g_{\text{para,syn}}(t) \mathcal{L}_{\text{para,syn}} \\ & + \lambda_{\text{para,cont}} g_{\text{para,cont}}(t) \mathcal{L}_{\text{para,cont}} \\ & + \lambda_{\text{ent}} g_{\text{ent}}(t) \mathcal{L}_{\text{ent}}. \end{aligned} \quad (3)$$

No content projection, learned router, extra corpus, or additional trainable module is enabled. The single-relation systems retain their original full-384 losses; Table 1 therefore summarizes implementation differences rather than a 2×2 factorial identity.

4 Experimental Setup

4.1 Data and Training

All configurations use the official BabyLM 2026 Strict-Small corpus released on Hugging Face¹ and the shared setup in Table 2 (Choshen et al., 2026). The corpus contains approximately 10M words. Models train for 10 epochs, with cumulative exposure no greater than 100M words. We evaluate seeds 0, 1, and 42 for every primary system. Seed 0 was designated as the submission seed before the final three-seed results were available, and it is also used in the matched checkpoint analyses. All systems have 34,677,952 parameters. Parameter-neutral means that the auxiliary objectives introduce no trainable parameters, although they do add target-construction and loss-computation cost.

4.2 Evaluation and Reporting

We evaluate models using the BabyLM 2026 evaluation pipeline² (Choshen et al., 2026). Conventional NLP Performance includes BLiMP, BLiMP

¹BabyLM 2026 Strict-Small

²Evaluation pipeline commit 3d57ddc8c40e.

Setting	Value
Backbone	DeBERTa-v2 (He et al., 2021), 12 layers, 12 heads
Hidden / FFN size	384 / 1280
Tokenizer	SentencePiece, 40,000 tokens
Parameters	34,677,952
Sequence length	64 (epochs 1–5); 256 (epochs 6–10)
MLM probability	Linear 0.40 $\rightarrow$ 0.15
Replacement	80% mask, 10% random, 10% unchanged
Optimizer	LAMB (You et al., 2020)
Peak learning rate	0.007
Schedule	1% warmup, cosine decay
Batch size	256

Table 2: Shared backbone and training configuration.

Supplement, EWoK, Entity Tracking, COMPS, GlobalPIQA, and (Super)GLUE, while Human-likeness and Development includes Reading and Age of Acquisition (AoA).

We follow the metrics implemented by this pipeline. Entity Tracking scores whether the correct continuation receives the highest probability. GlobalPIQA averages its parallel and non-parallel subsets. (Super)GLUE macro-averages BoolQ, MultiRC, RTE, WSC, MRPC, QQP, and MNLI; MRPC and QQP use binary F1, and the remaining tasks use accuracy. Reading is 100 times the mean normalized ΔR^2 for self-paced reading and eye tracking. AoA is 100 times the Pearson correlation between model-implied and child acquisition ages, with a score of zero when $p > 0.1$.

NLP Average is the arithmetic mean of the seven Conventional NLP metrics, Human Average is the arithmetic mean of Reading and AoA, and Overall is the unweighted arithmetic mean of all nine top-level metrics. Results are means with sample standard deviations over seeds 0, 1, and 42; comparisons with Standard MLM use same-seed paired differences. With three pairs, these differences are reported descriptively rather than as significance tests. Checkpoint analyses use matched runs with seed 0.

Method	BLiMP	B-Supp	EWoK	Entity	COMPS	GlobalPIQA	(Super)GLUE	NLP Avg
Standard MLM	66.95 $\pm$ 1.13	58.54 $\pm$ 1.07	50.83 $\pm$ 0.49	21.38 $\pm$ 1.02	51.96 $\pm$ 0.16	37.44 $\pm$ 2.05	65.06 $\pm$ 0.81	50.31 $\pm$ 0.13
Syntagmatic	66.88 $\pm$ 0.72	60.67 $\pm$ 1.46	50.69 $\pm$ 1.43	21.21 $\pm$ 0.40	52.19 $\pm$ 0.34	36.79 $\pm$ 2.34	65.78 $\pm$ 1.52	50.60 $\pm$ 0.29
Paradigmatic	67.14 $\pm$ 0.39	60.43 $\pm$ 1.66	51.59 $\pm$ 0.61	21.58 $\pm$ 1.05	52.22 $\pm$ 0.33	37.10 $\pm$ 2.16	65.63 $\pm$ 0.29	50.81 $\pm$ 0.27
Relay Joint	67.57 $\pm$ 1.18	61.56 $\pm$ 3.18	52.32 $\pm$ 0.75	20.51 $\pm$ 1.33	52.14 $\pm$ 0.08	34.34 $\pm$ 2.82	65.60 $\pm$ 0.77	50.58 $\pm$ 1.07

Table 3: Conventional NLP Performance, mean $\pm$ sample SD over three seeds. Bold marks the highest mean in each column.

5 Results

5.1 Overall Performance and Across-Seed Variation

Tables 3 and 4 show that the two single-relation systems are more consistent across seeds than Relay Joint. Paradigmatic has the highest mean Overall and exceeds its matched Standard MLM result in all three seeds. Syntagmatic also has a higher mean Overall, although its matched difference is positive in two of the three seeds. Relay Joint varies more: its submitted checkpoint trained with seed 0 is the strongest individual checkpoint among the four primary systems, but the advantage does not persist across seeds. The results therefore support the two relational targets, especially Paradigmatic, but do not show that the staged Relay integration is generally better.

5.2 Transfer Profiles and Integration Trade-offs

The overall averages hide clear differences across tasks. Relay Joint performs best on several grammar- and knowledge-related evaluations, whereas Paradigmatic provides the most consistent Overall improvement across the three seeds. No system improves every task family. The relational objectives therefore change which tasks benefit rather than producing uniform gains.

Table 5 shows a similar pattern across the integration variants. Clean Union has the highest mean NLP Average. It directly combines the unchanged single-relation recipes in the full 384-dimensional state, including their original activation gates, whereas Simultaneous Joint uses a separately specified simultaneous schedule. Adding routing, entity anchoring, or staging does not consistently improve performance. These comparisons therefore do not identify any component as necessary and do not show that the Relay schedule is generally preferable.

Method	Reading	AoA	Human Avg	Overall	Paired Δ vs. Std.	Positive seeds
Standard MLM	5.69 ± 0.39	0.00 ± 0.00	2.84 ± 0.19	39.76 ± 0.08	—	—
Syntagmatic	5.85 ± 0.57	0.00 ± 0.00	2.92 ± 0.28	40.01 ± 0.28	$+0.25 \pm 0.35$	2/3
Paradigmatic	6.00 ± 0.16	0.00 ± 0.00	3.00 ± 0.08	40.19 ± 0.21	$+0.43 \pm 0.18$	3/3
Relay Joint	5.94 ± 0.12	0.00 ± 0.00	2.97 ± 0.06	40.00 ± 0.82	$+0.24 \pm 0.88$	1/3

Table 4: Human-likeness and Overall, mean $\pm$ sample SD over three seeds. Paired differences are computed against the matched Standard MLM seed from unrounded Overall values. Overall is the unweighted arithmetic mean of all nine top-level metrics.

Configuration	Obj.	R	A	Ret.	Integration	NLP Avg
Standard MLM	None	No	No	No	—	50.31 ± 0.13
Simultaneous Joint	Both	No	No	No	Simultaneous	50.10 ± 0.75
Routing Only	Both	Yes	No	No	Simultaneous	50.60 ± 0.43
Entity Anchor Only	Both	No	Yes	No	Simultaneous	50.70 ± 0.92
Routing + Entity Anchor	Both	Yes	Yes	No	Simultaneous	50.65 ± 0.12
Clean Union	Both	No	No	No	Direct union	50.88 ± 0.37
Staged Joint	Both	Yes	Yes	No	Staged	50.55 ± 0.39
Relay Joint	Both	Yes	Yes	10%	Staged relay	50.58 ± 1.07

Table 5: Integration comparison over three seeds. Obj. indicates active relational objectives; R, A, and Ret. denote routing, entity anchoring, and Syntagmatic retention. NLP Average covers the seven Conventional NLP metrics and is not Overall. Staged Joint and Relay Joint both use relation-specific routing and entity anchoring; Relay Joint additionally retains the Syntagmatic objective at 10% strength after syntax-level Paradigmatic supervision begins.

5.3 Developmental Trajectories

Figure 2 separates the learning biases more clearly than it ranks the systems. The trajectories differ substantially across objectives, but no system dominates consistently across measures or exposure levels. Changes in relative position across checkpoints make the curves descriptive evidence of relation-

specific learning dynamics, not of uniformly accelerated development under one integration strategy.

These downstream results motivate a narrower question: whether the relational objectives leave distinguishable signatures in the learned representations even when aggregate transfer varies across seeds.

6 Representation Analysis

Downstream performance does not identify the mechanism behind a score change. We therefore use neighborhood retrieval to test the Paradigmatic signal and context-removal and dependency diagnostics to test the Syntagmatic signal. These probes measure target-related properties rather than general model quality; complete definitions and results appear in Appendix C.

6.1 Paradigmatic Neighborhood Structure

The probe asks whether a masked hidden state retrieves the correct ground truth token neighborhood among 64 candidates, using 2,000 shared items grouped by 31 target-token IDs. Figure 3 shows that all three relationally trained models move closer to the appropriate neighborhood in the full

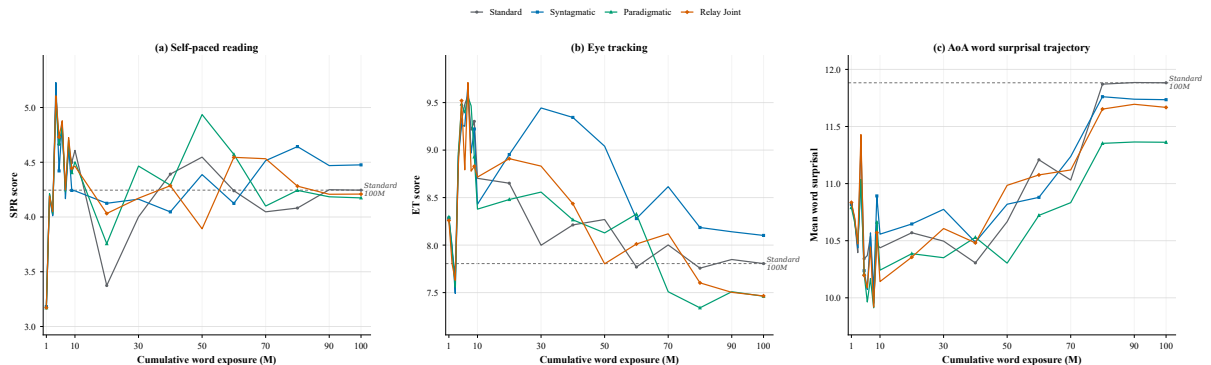


Figure 2: Matched Reading and AoA trajectories for seed 0. Higher is better for self-paced reading (SPR) and eye tracking (ET); lower is better for AoA target-word surprisal. Dashed lines mark the Standard MLM 100M reference. Curves are single runs and include no cross-run uncertainty bands. The AoA trajectory reports target-word surprisal and is distinct from the final AoA correlation score used in Table 4.

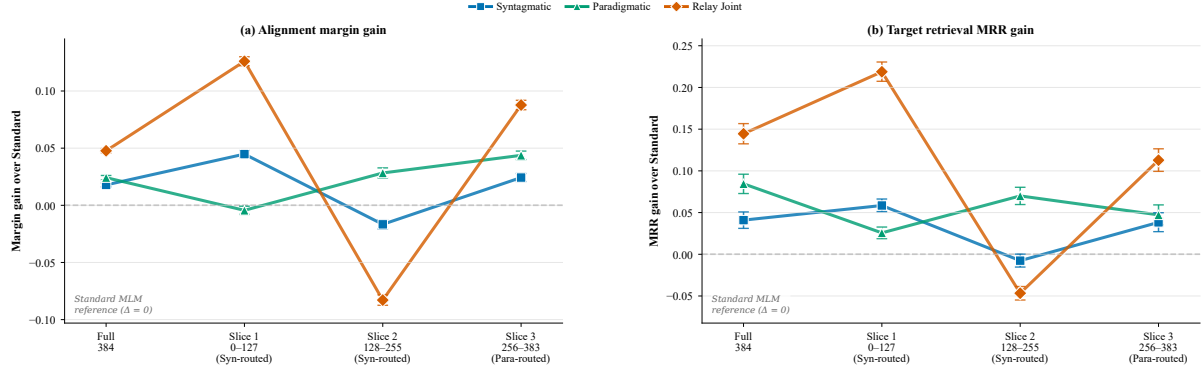


Figure 3: Paradigmatic neighborhood alignment for 2,000 shared items grouped into 31 target-token-ID clusters, reported as paired differences from Standard MLM. Points are item-macro paired means; bars are 95% intervals from 10,000 cluster-bootstrap resamples using target token ID as the resampling unit. Slices 1–2 cover dimensions 0–255 and Slice 3 covers dimensions 256–383. The intervals condition on matched checkpoints for seed 0 and do not measure variation across training seeds.

representation. Relay Joint has the largest change at the submitted checkpoint, improving alignment margin by +0.0477 and MRR by +0.1445 over Standard MLM. Both 95% cluster-bootstrap intervals remain above zero when target token ID is used as the resampling unit. These intervals describe the matched checkpoints for seed 0 only; they do not show that the same effect holds across training seeds.

6.2 Dimensionally Selective Reorganization

Relay Joint improves Slices 1 and 3 but reduces performance in Slice 2 relative to Standard MLM. Slices 1 and 2 both receive the Syntagmatic objective, yet they change in opposite directions. The hidden state therefore does not show a clean split between Syntagmatic and Paradigmatic information. Because these are fixed coordinate slices, the analysis also does not show that routing caused the observed pattern.

6.3 From Representation to Transfer

Removing attention-selected positions harms ground truth token prediction more than removing matched controls, indicating that selected context carries prediction-relevant information. The Syntagmatic model shows a larger average change, but with notable cross-seed fluctuation. Dependency recall shifts only slightly and precision is essentially unchanged. This is an independent Standard–Syntagmatic diagnostic, not mechanism evidence for Relay Joint.

The clearest representation and transfer results do not coincide. Paradigmatic gives the most consistent Overall gain across the three seeds, whereas

Relay Joint has the strongest neighborhood signature at the submitted checkpoint but variable downstream performance across seeds. Strong representational signals are therefore not sufficient to guarantee stable downstream generalization. These probes identify targeted internal properties relevant to the relational objectives and should not be treated as overall model-quality scores.

7 Limitations

Scope. We evaluate English BabyLM 2026 Strict-Small, one 34.7M-parameter DeBERTa-v2 architecture, and three seeds. Three seed-matched comparisons are insufficient for strong significance or broad generalization.

Relational approximations. Attention is not explicit syntax, embedding neighbors need not be genuine substitutes, and fixed English vocabulary categories are approximate. The fixed coordinate partition is basis-dependent and imposed rather than discovered.

Result boundaries. Directions vary across tasks and seeds; Relay Joint is positive against Standard for one of three seeds and trails Paradigmatic in mean Overall. The clustered bootstrap covers 31 syntax-token target IDs and conditions on one checkpoint per method. It controls repeated targets but not across-seed uncertainty or causality.

8 Conclusion

We introduced two self-generated relational targets without additional data, annotation, or trainable parameters. Across three seeds, both single-relation

systems yield higher mean Overall scores than Standard MLM, with Paradigmatic providing the highest and most consistent result. Relay Joint is less stable downstream but shows the strongest neighborhood alignment among the four primary systems at the submitted checkpoint. Its first and third slices improve while its second slice declines, so the results do not show a clean split between Syntagmatic and Paradigmatic information. Overall, relational supervision can help under limited exposure, but learning a relation and using it reliably for downstream tasks remain distinct challenges.

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A Training Configurations

Method	Syn onset	Syntax Para	Content Para	Representation	Anchor	Retention
Standard MLM	Off	Off	Off	–	Off	Off
Syntagmatic	~10%	Off	Off	Full 384	Off	Off
Paradigmatic	Off	~55%	final 1%	Full 384	Off	Off
Relay Joint	~10%	~20%	final 1%	Syn 256 / Para 128	~10–60%	10% Syn

Table 6: Configuration specification. All systems have 34,677,952 parameters. Routing uses fixed coordinates, not learned projections.

Both targets use $K = 8$. Syntagmatic uses the last attention layer with weight 5×10^{-4} . Syntax Paradigmatic uses 5×10^{-5} ; content Paradigmatic uses 10^{-6} . Relay Joint’s full-state anchor has weight 2.5×10^{-4} . No relational component adds trainable parameters.

Objective	Eligible MLM-selected positions
Syntagmatic	Syntax, reading-sensitive, or content category; entity and invalid tokens are excluded.
Paradigmatic, syntax	Fixed syntax-list match; active after about 55% alone and about 20% in Relay Joint.
Paradigmatic, content	Content category; active only in the final 1%. Entity and reading-sensitive categories have zero weight.
Entity anchor	Relay Joint entity category; active from about 10% through 60%.

Table 7: Ground truth token eligibility. Categories are mutually exclusive fixed token-ID lookups built before training.

Vocabulary-derived token categories. No POS tagger or NER system is used. The tokenizer vocabulary is scanned once before training. One recognized leading SentencePiece, byte-level BPE, WordPiece, or space marker is removed; trailing punctuation is stripped; and fixed-list matching is case-insensitive. Byte tokens and entries with no alphabetic character are invalid. The lookup then assigns the first matching category in this order: syntax, reading-sensitive, entity, and content. Thus the categories are mutually exclusive even when lexical lists overlap. An entity is a remaining entry containing a digit, or an alphabetic entry of at least three characters that begins with an uppercase letter. Remaining alphabetic entries are content tokens.

Fixed lexical lists. **Auxiliaries:** is, are, was, were, be, been, being, am, do, does, did, have, has, had, can, could, will, would, should, may, might, must, shall, ought, need, dare, used. **Determiners:** a, an, the, this, that, these, those, some, any, all, each, every, many, few, no, much, more, most, several, both, either, neither, such, what. **Pronouns:** he, she, it, they, him, her, them, his, their, its, we, us, our, you, your, i, me, my, myself, yourself, himself, herself, itself, ourselves, themselves, one, ones. **Negation:** not, n't, never, no, nor, neither. **Prepositions:** in, on, at, by, with, from, to, of, for, into, onto, over, under, near, about, between, through, during, without, within, along, across, behind, beyond, toward, towards, upon, among, amongst, beside, besides, against, around, before, after, above, below, off, up, down, out. **Conjunctions:** and, or, but, because, although, if, when, while, before, after, since, until, unless, whereas, so, yet, than, as, though, whether, once, till, lest, except, provided, given, suppose, assuming, whenever, wherever. **Wh-words:** who, whom, whose, which, where, why, how, what, whatever, whichever, whoever, however, wherever. **Comparatives:** more, less, fewer, better, worse, bigger, smaller, higher, lower, longer, shorter, older, younger. **Temporal expressions:** now, then, ago, later, earlier, soon, already, still, yet, finally, eventually, previously, formerly, currently, recently, lately, immediately, suddenly, gradually.

Ground truth token and context handling. Target-position eligibility is looked up from the clean ground truth token ID, never the corrupted input. The same category therefore applies whether an MLM-selected token is replaced with [MASK], a random token, or left unchanged. The Syntagmatic context set excludes padding, [CLS], [SEP], [MASK], the target itself, and every other MLM-selected position. Context embeddings are then gathered from the original uncorrupted token IDs. Subword entries are classified independently; the code performs no word reconstruction.

B Per-seed and Component Results

Method	Seed	BLiMP	B-Supp	EWoK	Entity	COMPS	GlobalPIQA	(S)GLUE	Reading	AoA	Overall
Standard MLM	0	66.80	58.57	51.39	22.53	51.90	35.13	64.82	6.03	0.00	39.69
Standard MLM	1	68.15	57.46	50.60	21.01	52.14	39.06	64.39	5.76	0.00	39.84
Standard MLM	42	65.90	59.60	50.49	20.59	51.85	38.11	65.96	5.27	0.00	39.75
Syntagmatic	0	67.56	60.37	49.10	20.84	52.50	38.62	67.52	6.29	0.00	40.31
Syntagmatic	1	66.12	59.38	51.10	21.64	51.82	37.61	65.00	5.21	0.00	39.76
Syntagmatic	42	66.97	62.26	51.88	21.16	52.26	34.15	64.81	6.04	0.00	39.95
Paradigmatic	0	67.13	62.30	52.28	21.67	52.22	34.62	65.78	5.81	0.00	40.20
Paradigmatic	1	67.53	59.89	51.14	22.59	51.89	38.58	65.81	6.11	0.00	40.39
Paradigmatic	42	66.76	59.11	51.35	20.49	52.55	38.09	65.30	6.09	0.00	39.97
Relay Joint	0	67.84	65.21	53.00	21.47	52.24	37.59	65.29	5.83	0.00	40.94
Relay Joint	1	66.28	60.09	52.44	21.07	52.09	32.71	65.03	5.92	0.00	39.51
Relay Joint	42	68.59	59.37	51.51	18.99	52.10	32.71	66.47	6.07	0.00	39.53

Table 8: Primary-system results by seed. Overall is the unweighted arithmetic mean of all nine top-level metrics.

Configuration	Seven-metric NLP Average
Standard MLM	50.31 $\pm$ 0.13
Syntagmatic	50.60 $\pm$ 0.29
Paradigmatic	50.81 $\pm$ 0.27
Simultaneous Joint	50.10 $\pm$ 0.75
Routing Only	50.60 $\pm$ 0.43
Entity Anchor Only	50.70 $\pm$ 0.92
Routing + Entity Anchor	50.65 $\pm$ 0.12
Clean Union	50.88 $\pm$ 0.37
Staged Joint	50.55 $\pm$ 0.39
Relay Joint	50.58 $\pm$ 1.07

Table 9: All configurations, mean $\pm$ sample SD over three seeds. This comparable aggregate excludes Reading and AoA and is not Overall. Clean Union directly combines full-state objectives without routing or anchoring.

C Representation Analyses

Paradigmatic neighborhood probe. The probe uses 2,000 shared syntax-token instances and 31 target-token IDs, used as bootstrap clusters, retrieving the correct eight-neighbor centroid among 64 candidates. Table 10 gives all paired item means and target-token cluster intervals from 10,000 resamples.

Table 10: All neighborhood-probe comparisons. Intervals are 95% target-token clustered-bootstrap intervals conditional on matched checkpoints for seed 0, not across-seed intervals.

Comparison	Space	Metric	Mean	CI low	CI high
Syn.–Std.	Full	Margin	+0.0178	+0.0114	+0.0227
Syn.–Std.	Full	MRR	+0.0409	+0.0058	+0.0796
Syn.–Std.	S1	Margin	+0.0448	+0.0333	+0.0522
Syn.–Std.	S1	MRR	+0.0585	+0.0380	+0.0797
Syn.–Std.	S2	Margin	-0.0166	-0.0258	-0.0047
Syn.–Std.	S2	MRR	-0.0076	-0.0262	+0.0118
Syn.–Std.	S3	Margin	+0.0243	+0.0124	+0.0325
Syn.–Std.	S3	MRR	+0.0383	+0.0137	+0.0638
Para.–Std.	Full	Margin	+0.0242	+0.0180	+0.0318
Para.–Std.	Full	MRR	+0.0845	+0.0388	+0.1463
Para.–Std.	S1	Margin	-0.0044	-0.0207	+0.0139
Para.–Std.	S1	MRR	+0.0257	+0.0033	+0.0625
Para.–Std.	S2	Margin	+0.0283	+0.0129	+0.0406
Para.–Std.	S2	MRR	+0.0700	+0.0395	+0.1197
Para.–Std.	S3	Margin	+0.0437	+0.0355	+0.0541

Comparison	Space	Metric	Mean	CI low	CI high
Para.–Std.	S3	MRR	+0.0472	+0.0233	+0.0794
Relay–Std.	Full	Margin	+0.0477	+0.0358	+0.0566
Relay–Std.	Full	MRR	+0.1445	+0.0773	+0.1977
Relay–Std.	S1	Margin	+0.1260	+0.1031	+0.1420
Relay–Std.	S1	MRR	+0.2188	+0.1482	+0.2812
Relay–Std.	S2	Margin	-0.0829	-0.1013	-0.0679
Relay–Std.	S2	MRR	-0.0467	-0.0794	-0.0238
Relay–Std.	S3	Margin	+0.0877	+0.0741	+0.0982
Relay–Std.	S3	MRR	+0.1128	+0.0774	+0.1363
Relay–Syn.	Full	Margin	+0.0299	+0.0211	+0.0368
Relay–Syn.	Full	MRR	+0.1036	+0.0463	+0.1517
Relay–Syn.	S1	Margin	+0.0812	+0.0641	+0.0935
Relay–Syn.	S1	MRR	+0.1603	+0.0985	+0.2146
Relay–Syn.	S2	Margin	-0.0663	-0.0873	-0.0501
Relay–Syn.	S2	MRR	-0.0390	-0.0645	-0.0200
Relay–Syn.	S3	Margin	+0.0634	+0.0522	+0.0757
Relay–Syn.	S3	MRR	+0.0745	+0.0361	+0.1065
Relay–Para.	Full	Margin	+0.0235	+0.0115	+0.0335
Relay–Para.	Full	MRR	+0.0600	+0.0056	+0.1060
Relay–Para.	S1	Margin	+0.1304	+0.1073	+0.1483
Relay–Para.	S1	MRR	+0.1931	+0.1237	+0.2587
Relay–Para.	S2	Margin	-0.1112	-0.1206	-0.1022
Relay–Para.	S2	MRR	-0.1166	-0.1723	-0.0821
Relay–Para.	S3	Margin	+0.0440	+0.0299	+0.0531
Relay–Para.	S3	MRR	+0.0656	+0.0249	+0.0913

The largest cluster is the token a (448 items). Cluster intervals average 3.74 times the width of item intervals. Full-384 is the primary analysis. The three 128-dimensional diagnostic slices are coordinate-based analyses, not discovered modules.

Syntagmatic diagnostics. Attention is ranked over SentencePiece positions before word-level mapping. Selected subwords are mapped to Universal Dependencies words, with duplicate words collapsed. Recall@8 is the fraction of ground truth direct dependency-neighbor words recovered; Precision@8 is the fraction of unique mapped selected words that are direct dependency neighbors. For context removal, the same selected token positions are zeroed in the input to the final Transformer layer. A matched control removes the same number of random non-special, non-target positions. Tables 11 and 12 include only the current Standard MLM and Syntagmatic runs.

Method	Seed / summary	Recall@8	Precision@8
Standard MLM	0 / 1 / 42	.8707 / .8806 / .8699	.2997 / .3003 / .2970
	Mean $\pm$ SD	.8737 $\pm$.0059	.2990 $\pm$.0018
Syntagmatic	0 / 1 / 42	.8750 / .8759 / .8772	.2999 / .2979 / .2987
	Mean $\pm$ SD	.8760 $\pm$.0011	.2988 $\pm$.0010

Table 11: Dependency alignment by seed and sample mean $\pm$ SD.

Method	Seed / summary	Attended	Matched	Difference
Standard MLM	0 / 1 / 42	1.6525 / 1.6137 / .9038	-.0447 / .1678 / -.0039	1.6972 / 1.4459 / .9077
	Mean $\pm$ SD	1.3900 $\pm$.4215	.0397 $\pm$.1128	1.3503 $\pm$.4034
Syntagmatic	0 / 1 / 42	1.8058 / 2.1002 / .7960	-.0438 / .0159 / -.0082	1.8496 / 2.0842 / .8042
	Mean $\pm$ SD	1.5673 $\pm$.6840	-.0120 $\pm$.0300	1.5793 $\pm$.6814

Table 12: Ground truth token NLL increase after context removal, by seed and sample mean $\pm$ SD. Difference is attended minus matched.